

Advanced Network Security Features



Advanced Network Security Features

- Zero Trust
- SASE
- SWG
- DLP
- Authentication
- MFA concepts

Labs

- Testing the DLP Policy
- Exploring Advanced features of Palo Alto Firewall

Zero Trust

- Zero Trust is a modern cybersecurity framework that operates on a core principle:

Never Trust, Always Verify

- every user, device, application, and network flow must be continuously authenticated, authorized, and validated before being granted access.
- Zero Trust assumes:
 - No network (internal or external) is inherently safe.
 - Identities and devices can be compromised at any time.
 - Lateral movement must be minimized.
 - Access must be precisely scoped and continuously monitored.
- It shifts security from perimeter-based to identity-, device-, and context-based access control.



Core Principles of Zero Trust

Verify Explicitly

Access decisions must consider:

- User identity
- Device health/status
- Location
- Application type
- Data classification
- Behavioral analytics

No implicit trust based on network location or historical access.

Least Privilege Access

Grant minimum required permissions:

- Just-in-Time (JIT) access
- Just-Enough-Access (JEA)
- Micro-segmentation
- Privileged Access Management (PAM)

Assume Breach

Architect assuming attackers are already inside:

- Monitor internal traffic
- Strict segmentation
- Rapid detection and response
- Forensic visibility across endpoints, network, and cloud

Zero Trust Architecture (ZTA) Components

A mature Zero Trust model covers identity, devices, networks, workloads, data, and applications.

Identity

- Strong authentication using MFA
- Conditional access policies
- Continuous identity risk scoring
- Privileged Identity Management (PIM)

Devices

- Device posture validation (patched, compliant, encrypted)
- MDM/UEM integration
- Endpoint Detection and Response (EDR)
- Non-compliant devices are quarantined

Network

- Micro-segmentation
- Identity-aware firewall policies
- Zero Trust Network Access (ZTNA) replacing VPN
- East-West traffic inspection



Zero Trust Architecture (ZTA) Components

Applications

- Application-aware access controls
- API gateways with token validation
- Secure software supply chain
- Runtime protection (RASP, WAF, CSPM)

Data

- Data classification
- DLP enforcement
- Encryption at rest and in transit
- Access control tied to identity + sensitivity

Workloads (On-Prem + Cloud)

- Container security
- Cloud posture management
- Threat modelling for microservices
- Zero Trust for hybrid environments



Zero Trust Flow (How It Works)

A typical Zero Trust access workflow:

1. User attempts access
2. Identity verification through MFA
3. Device posture validated
4. Context evaluated (location, risk, behavior)
5. Access policy engine evaluates rules
6. Least privilege access is granted
7. Continuous monitoring (UEBA, EDR)
8. Real-time policy re-evaluation if risk changes
9. Session terminated or restricted if anomalies are detected



Zero Trust Technologies

Access & Identity

- MFA
- Identity Providers (Azure AD, Okta, Ping)
- Conditional Access
- PAM/PIM

Network & Transport

- Micro-segmentation (Illumio, Cisco, VMware NSX)
- ZTNA solutions (Zscaler, Palo Alto Prisma, Cloudflare Zero Trust)
- NGFW with identity integration

Endpoint

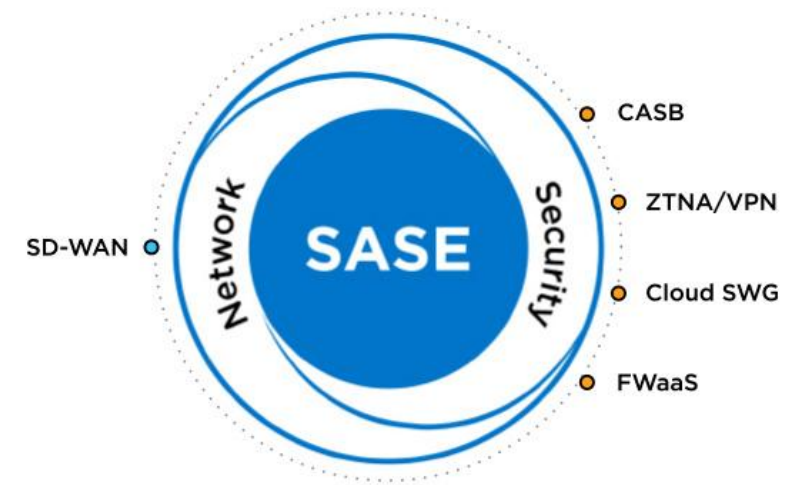
- EDR/XDR platforms
- MDM/UEM (Intune, Workspace ONE, MobileIron)

Visibility & Monitoring

- SIEM + SOAR
- UEBA
- DNS filtering
- Deception technologies

Secure Access Service Edge (SASE)

- SASE (pronounced “sassy”) is a cloud-delivered cybersecurity and networking architecture introduced by Gartner. It converges networking and security into a unified, cloud-native service.
- SASE integrates:
 - ✓ SD-WAN (Software-Defined WAN)
 - ✓ Security Service Edge (SSE)
 - ✓ Zero Trust Network Access (ZTNA)
 - ✓ Cloud-delivered security controls
- The aim is to provide secure, *optimized access to applications from anywhere*, whether users connect from branches, home offices, or mobile devices.
- SASE delivers networking + security as a *single, centrally managed, cloud-distributed service*.



Why SASE?

- **Remote Workforce**

- Employees connect from anywhere; perimeter-based firewalls cannot scale.

- **Cloud-First Adoption**

- Applications are now in SaaS, IaaS, and multi-cloud environments.
- Traffic no longer flows through data centers.

- **Legacy Network Limitations**

- MPLS, on-prem firewalls, and VPNs introduce:
 - High latency
 - Backhauling costs
 - Complexity in managing multiple appliances

- **Unified Governance**

- SASE centralizes security policies and monitoring across distributed environments.

Core Components of SASE

SASE = **SD-WAN** + **SSE** (Security Service Edge)

SD-WAN (Network Component)

- Delivers:
 - Intelligent routing
 - Application-aware traffic steering
 - Bandwidth optimization
 - Transport independence (MPLS, broadband, LTE/5G)
- Eliminates costly backhauling to the data centre.



Core Components of SASE

Security Service Edge (SSE)

- Cloud-delivered security that enforces Zero Trust principles. SSE includes:

Zero Trust Network Access (ZTNA)

- Replaces traditional VPN
- Identity + device posture validation
- Provides least privilege, application-level access

Secure Web Gateway (SWG)

- URL filtering
- Malware protection
- Enforces acceptable use policies

Firewall-as-a-Service (FWaaS)

- Cloud-hosted full firewall stack
- Intrusion detection and prevention (IPS)
- Layer 3–7 inspection

Cloud Access Security Broker (CASB)

- Controls access to SaaS applications
- Data Loss Prevention (DLP)
- Shadow IT detection
- API-based SaaS security

DNS Security

- Blocks malicious domains
- Enforces DNS-layer filtering
- Together, SSE provides full cloud-native threat protection

Architecture of SASE

High-Level Architecture

- Users connect to the nearest SASE Point of Presence (PoP)
- PoPs provide both networking acceleration and security inspection
- Traffic is routed directly to cloud apps, SaaS, or data center
- Policies are centrally managed and applied globally

Traffic Flow

- User/device connects to the closest SASE PoP
- Identity verification + posture checks occur
- Security stack (ZTNA, FWaaS, CASB, SWG) inspects traffic
- SD-WAN optimizes routing
- Traffic proceeds to the destination with minimal latency

Benefits of SASE

- **Improved User Experience**

- Direct cloud access reduces latency.

- **Consistent Security**

- Unified policy enforcement, regardless of:
 - Location
 - Device
 - Network

- **Reduced Costs**

- Less dependency on MPLS and on-prem appliances.

- **Simplified Operations**

- Single platform for:
 - Networking
 - Security
 - Monitoring

- **Enhanced Security Posture**

- Zero Trust access model reduces attack surface.

SASE Deployment Roadmap

Phase 1: Assess

- Evaluate existing WAN, VPN, and security stack
- Identify cloud, SaaS, remote access patterns

Phase 2: Start with SSE

- Deploy ZTNA to replace VPN
- Introduce SWG, CASB, and FWaaS

Phase 3: SD-WAN Modernization

- Replace MPLS with SD-WAN gradually
- Local internet breakouts via PoPs

Phase 4: Consolidate Policies

- Create unified access and security policy framework

Phase 5: Optimize + Monitor

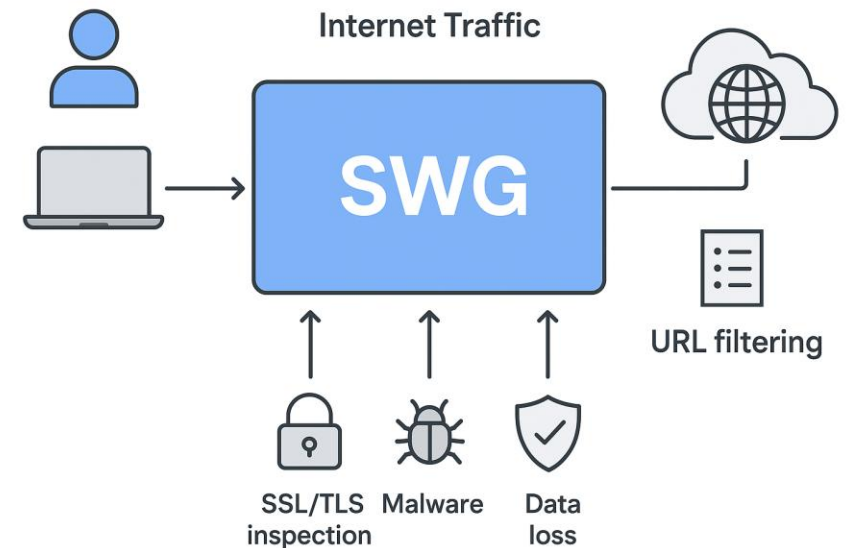
- Integrate with SIEM/SOAR
- Analyse logs for usage, anomalies, experience metrics

SASE vs. SSE

Component	SASE	SSE
Includes SD-WAN	Yes	No
Focus	Networking + Security	Security-only
Ideal For	Full architecture transformation	Initial Zero Trust security

Secure Web Gateway (SWG)

- A Secure Web Gateway (SWG) is a cybersecurity solution that protects users from web-based threats and enforces organizational web access policies.
- It acts as a control point between users and the internet, inspecting and filtering all outbound and inbound web traffic.
- SWG ensures:
 - Safe access to websites
 - Protection against malware, phishing, and malicious URLs
 - Enforced compliance with business policies
 - Monitoring and control over user web activity
- SWGs are essential for securing remote users, branches, and cloud-first environments.



Why SWG Is Needed

- Modern enterprises face challenges such as:
 - Phishing and malicious websites
 - Malware downloads from the web
 - Shadow IT (unsanctioned cloud applications)
 - Data leakage via web channels
 - Workforce accessing internet from remote/non-corporate networks
- SWGs provide a centrally managed control layer to protect users and data regardless of their network location.



Core Capabilities of SWG

- **URL Filtering**

- Custom allow/deny list.
- Blocks categories (like gambling, adult content, social media).

- **SSL/TLS Inspection**

- Decrypts HTTPS traffic
- Inspects content for threat

- **Malware and Threat Protection**

- Anti-virus and anti-malware scanning
- Sanding suspicious downloads.

Core Capabilities of SWG

- **Application Control**

- Controls apps like Google Drive, Dropbox
- Web-based tools or social media platforms.

- **Data Loss Prevention (DLP)**

- Prevents sensitive data from leaving the organization:
 - Credit card number
 - Business documents
 - Source code

- **Policy Enforcement**

- Centralized policies based on:
 - User identity, group or department
 - Device posture, location

- **Reporting and Analytics**

- Real-time dashboards, bandwidth usage
- Threat metrics, compliance reports

Types of SWG Deployments

- **Cloud SWG**

- Delivered via global PoPs. Ideal for:
- Remote users
- Hybrid workforce
- Cloud-first environments
- Examples: Zscaler, Netskope, Cisco Umbrella, Cloudflare

- **Hybrid SWG**

- Combination of cloud + on-prem for large distributed enterprises.

- **On-Premise SWG**

- Traditional appliance or virtual appliance:
- Deployed at data centres
- Suitable for legacy networks
- Examples: Palo Alto, Fortinet, Blue Coat

Role of SWG in SASE and Zero Trust

▪ In SASE

- SWG is a key component of SSE (Security Service Edge).
- It integrates with:
 - ZTNA
 - CASB
 - FWaaS
- A unified cloud SWG ensures every web request is inspected, regardless of user location.

▪ In Zero Trust

- SWG enforces:
 - Continuous verification
 - Least privilege access to web resources
 - Risk-based restrictions
- SWG is often the frontline control for enforcing zero trust at the web layer.

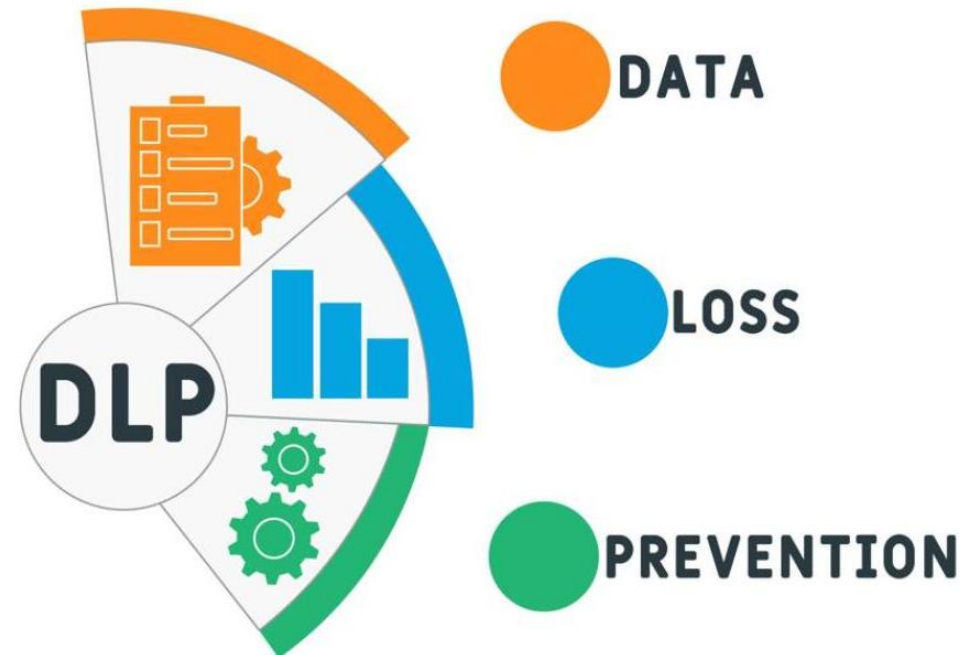
Benefits of SWG

Benefit	Description
Protection Against Web Threats	Blocks malware, phishing, drive-by downloads.
Policy Enforcement	Ensures compliance with corporate usage guidelines.
Cloud Application Visibility	Detects shadow IT and unsanctioned app usage.
Data Protection	Prevents data exfiltration via web channels.
Enhanced User Experience	Cloud SWGs eliminate backhauling, reducing latency.
Uniform Protection	Same policy applies to office, remote, and mobile users.



Data Loss Prevention (DLP)

- DLP is a cybersecurity discipline focused on preventing unauthorized exposure, transfer, or destruction of sensitive data.
- DLP ensures that data—whether in use, in motion, or at rest—is properly controlled, monitored, and protected according to organizational policies and regulatory requirements.
- DLP is used to protect:
 - Personally identifiable information (PII)
 - Payment card information (PCI)
 - Intellectual property (IP)
 - Financial data
 - Confidential business information
 - Source code and design documents



Objectives of DLP

Prevent Data Leakage

Stops sensitive data from leaving the organization via:

- Email
- Web uploads
- Cloud apps
- Removable media
- Printing

Enforce Compliance

Meets requirements for:

- GDPR
- HIPAA
- PCI-DSS
- SOX
- ISO 27001
- RBI/SEBI guidelines (India)

Visibility

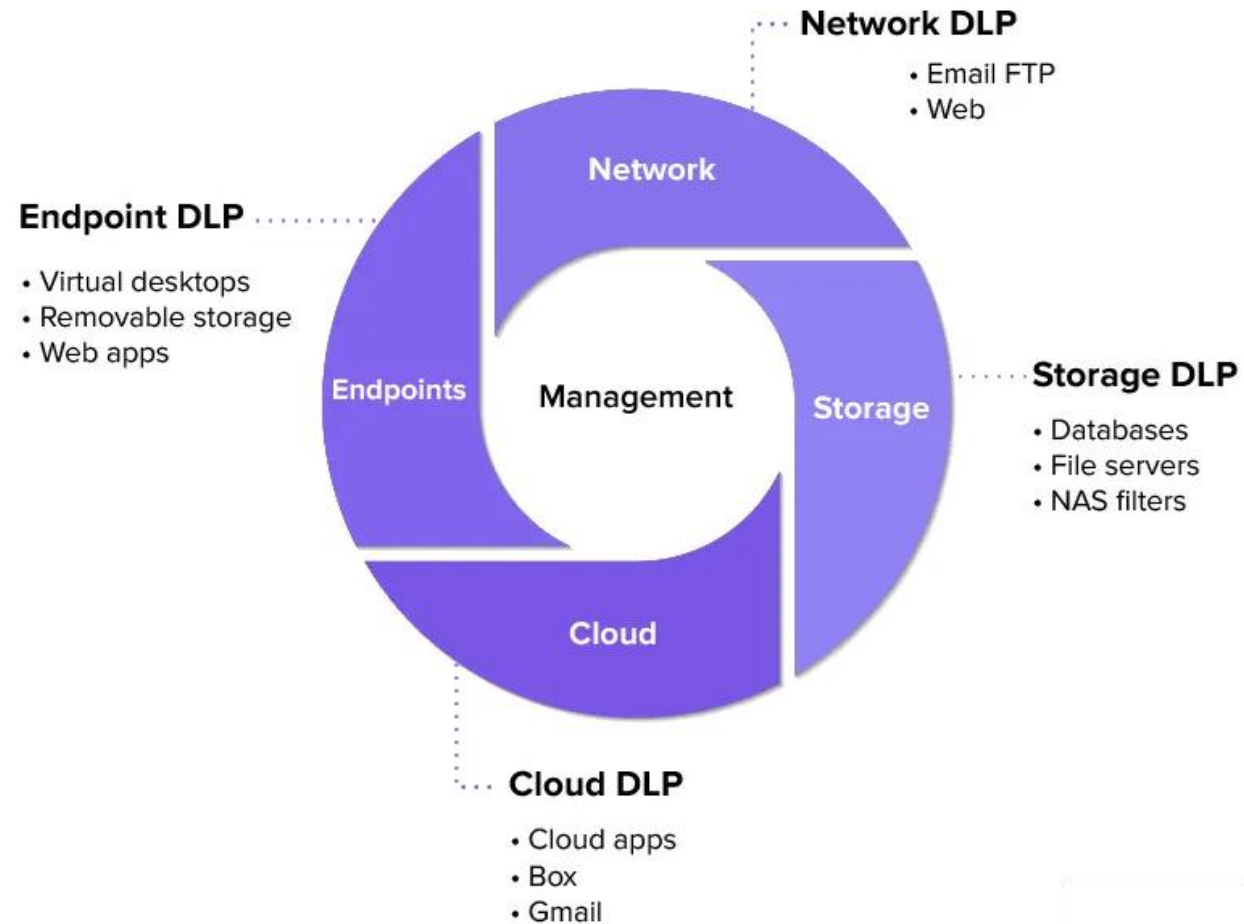
Provides insight into:

- Where sensitive data resides
- Who accesses it
- How it is shared

Data Governance

Ensures data is classified, handled, and used properly.

Types of DLP Solutions



Types of DLP Solutions

- **Network DLP**

- Deployed at perimeter or cloud proxy.
- Monitors outbound traffic
- Email scanning
- Web uploads control
- API-level cloud integration

- **Endpoint DLP**

- Agents on laptops/desktops.
- Controls USB/clipboard/printing
- Blocks unauthorized apps
- Inspects local files

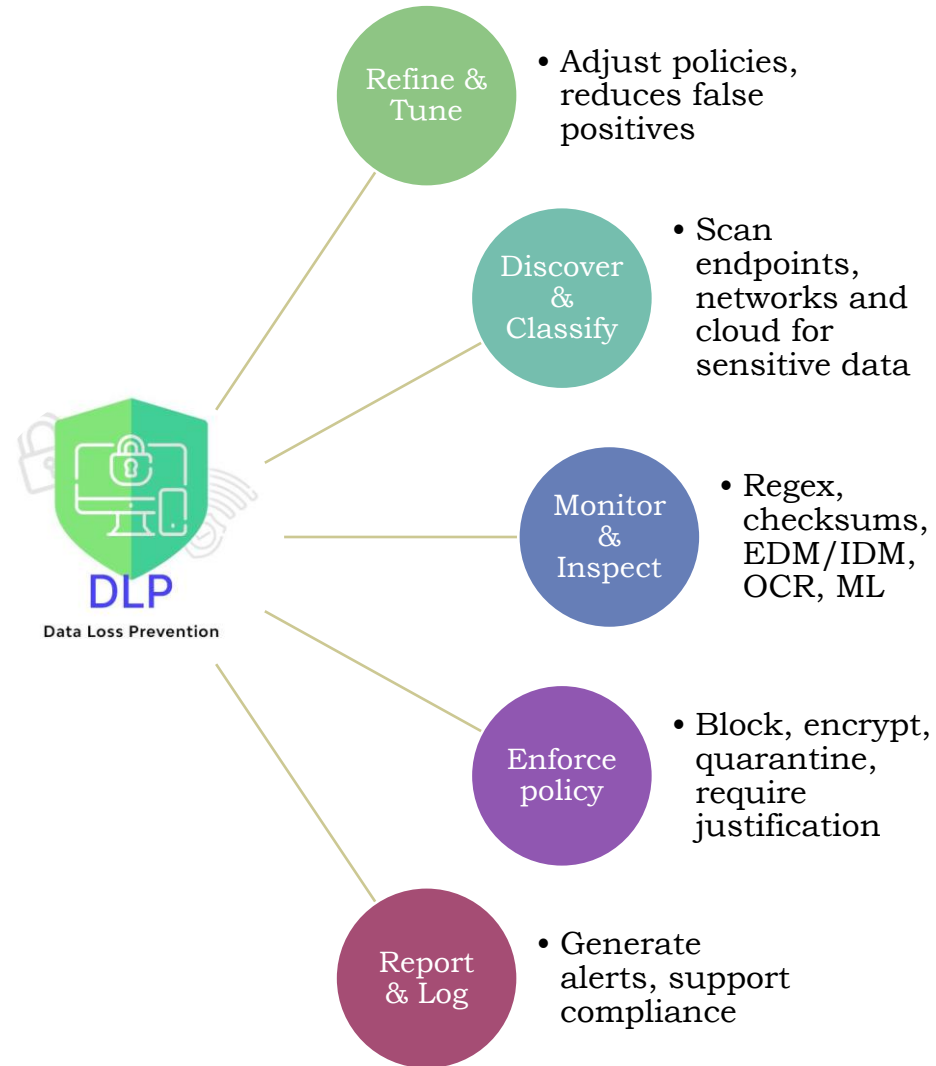
- **Cloud DLP**

- Delivered through CASB or SaaS.
- Protects data in SaaS applications
- Enforces data classification and sharing rules
- Provides real-time user activity monitoring

- **Storage/Discovery DLP**

- Scans on-prem and cloud storage repositories for:
 - Unencrypted sensitive data
 - Misplaced files with PII
 - Overly permissive access rights

How DLP works?



Categorization Of Data Loss Prevention

The Data Loss prevention can be classified into:

- **The Standard Measure**

- The standard security such as anti-virus software, intrusion detection system, firewall.
- These products can guide computers against external threats or intrusions.

- **The Advanced Measure**

- The Advanced measure employs the machine learning, and calculations to identify irregular access to data.
- This makes a difference to distinguish an unusual mail trade inside the framework.

- **The Designated System**

- The Designated System helps to detect and quickly prevent any unauthorized attempts to send, share or copy sensitive company data either intentionally or unintentionally.



Authentication

- **Authentication** is the cybersecurity process of verifying that a user, device, application, or system is **who or what it claims to be** before granting access.
- It is the first pillar of access control, followed by:
 - **Authorization** – What are you allowed to do?
 - **Accounting/Auditing** – What did you do?
 - Together, they form the AAA security model.



Why Authentication?

- Authentication prevents:
 - Unauthorized access
 - Credential theft misuse
 - Account takeover (ATO)
 - Lateral movement inside networks
- Weak authentication is a leading cause of breaches, especially in:
 - Phishing attacks
 - Credential stuffing
 - Brute-force attacks



Authentication Factors

- **Something You Know**

- Password
- PIN
- Security questions
- Weakness: Easily stolen, guessed, or phished.

- **Something You Have**

- Smart card
- Hardware token
- OTP app
- SMS OTP (less secure)
- Strength: Requires physical possession.

- **Something You Are (Biometrics)**

- Fingerprint
- Face recognition
- Iris scan
- Voice recognition
- Strength: Hard to replicate
- Risk: Privacy and irrevocability issues.

- **Something You Do**

- Typing rhythm
- Mouse movement patterns
- Used in continuous and adaptive authentication.

- **Somewhere You Are**

- Location-based authentication
- IP address, GPS
- Often used as a contextual factor.

Types of Authentication

- **Single-Factor Authentication (SFA)**

- Uses one factor (usually a password).
- Risk: High compromise probability.

- **Multi-Factor Authentication (MFA)**

- Uses two or more independent factors:
 - Password + OTP
 - Smart card + PIN
 - Biometrics + token
- Industry standard for enterprise security.

- **Two-Factor Authentication (2FA)**

- A subset of MFA using exactly two factors.

- **Passwordless Authentication**

- Eliminates passwords using:
 - Biometrics
 - FIDO2 security keys
 - Certificates
- Reduces phishing and credential theft.

Authentication Protocols and Standards

Protocol	Purpose
Kerberos	Secure authentication in Active Directory
LDAP	Directory-based authentication
RADIUS	Network access authentication
TACACS+	Device admin authentication
SAML	Web-based SSO
OAuth 2.0	Authorization delegation
OpenID Connect (OIDC)	Identity authentication on OAuth
FIDO2 / WebAuthn	Passwordless authentication



Authentication in Enterprise Environments

- **User Authentication**

- Corporate login (AD / Azure AD)
- MFA enforcement
- Conditional access policies

- **Device Authentication**

- Device certificates
- TPM-based authentication
- MDM/UEM validation

- **Application Authentication**

- API keys
- OAuth tokens
- Mutual TLS (mTLS)

- **Machine-to-Machine Authentication**

- Certificates
- Service principals
- Managed identities

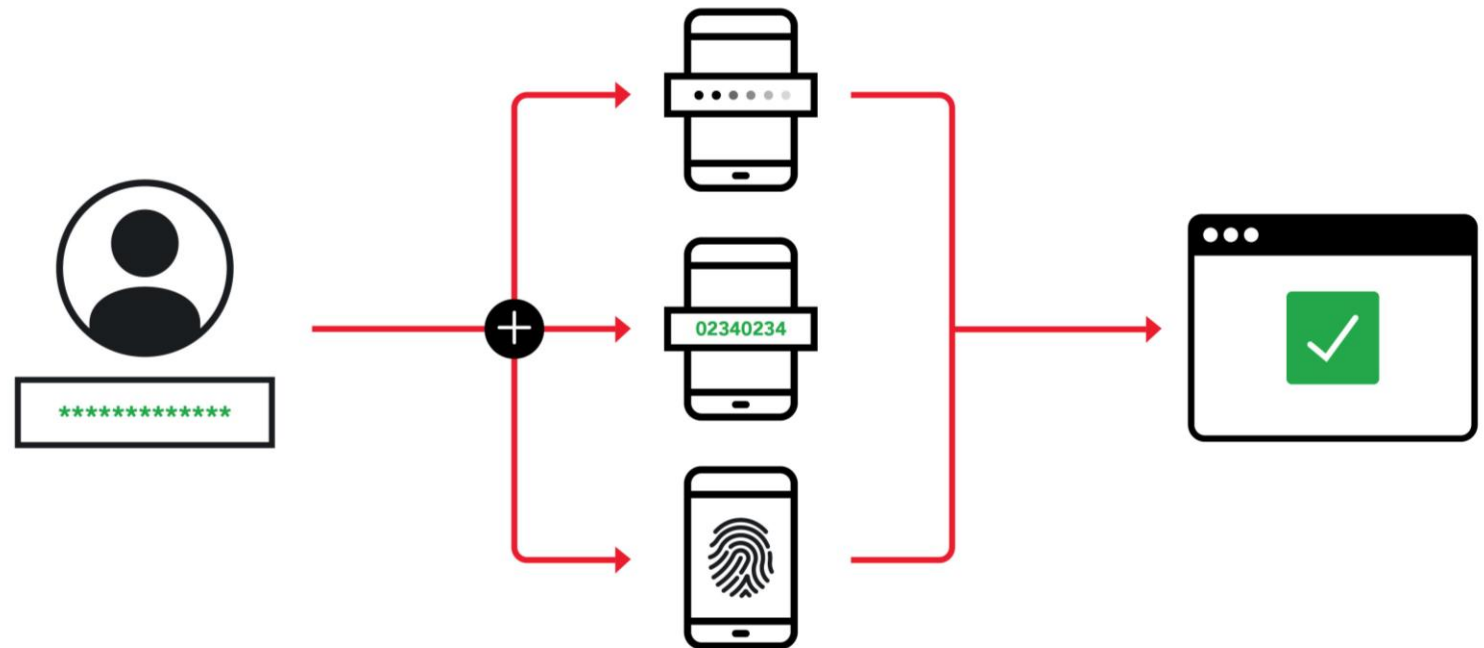
Common Authentication Attacks

- Phishing
- Credential stuffing
- Brute-force attacks
- Pass-the-hash
- MFA fatigue attacks
- Mitigations:
 - Strong MFA
 - Passwordless authentication
 - Behavioural analytics
 - Rate limiting

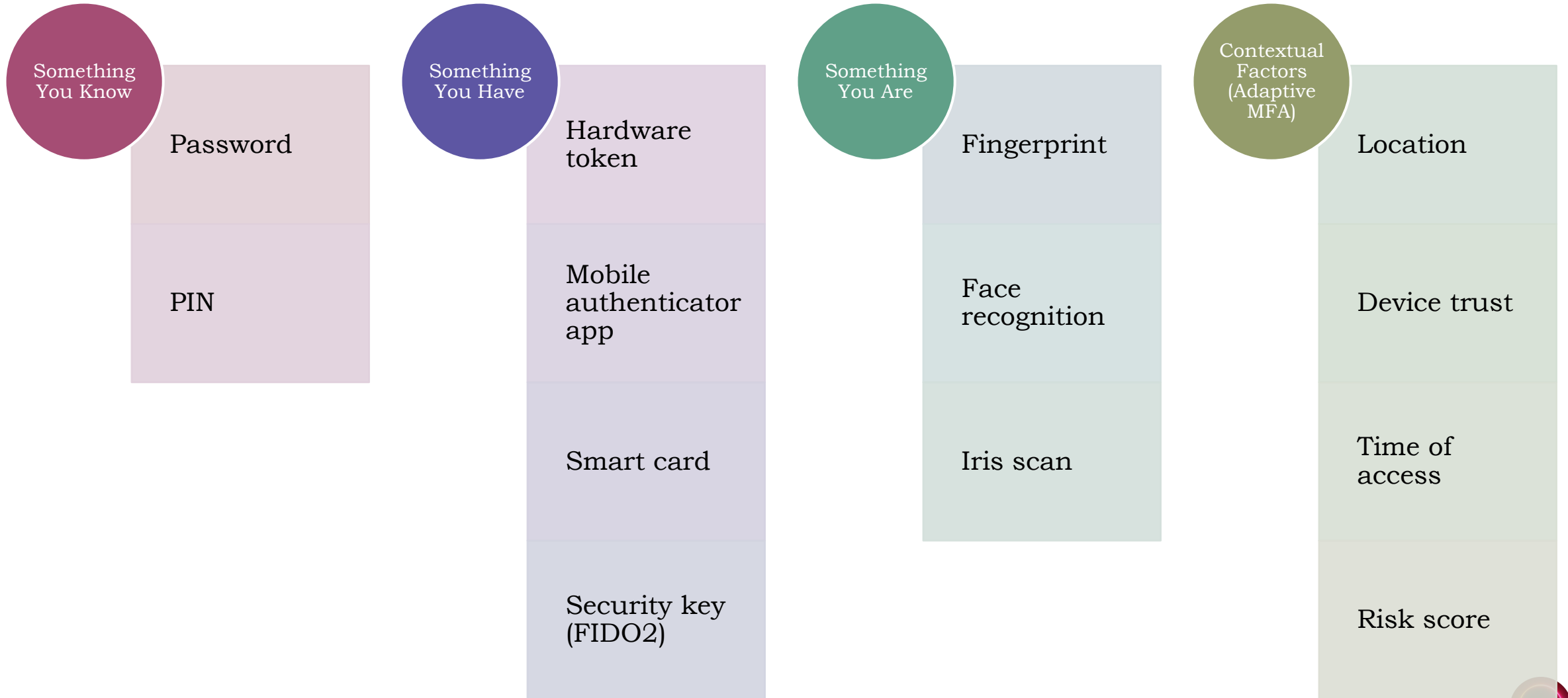


Multi-Factor Authentication (MFA)

- Multi-Factor Authentication (MFA) is a security mechanism that requires two or more independent authentication factors to verify a user's identity before granting access.
- MFA significantly reduces the risk of:
 - Phishing
 - Credential theft
 - Password reuse attacks
 - Account takeover (ATO)



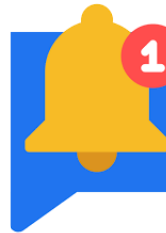
MFA Authentication Factors



MFA Authentication Methods



OTP (One-Time Password)



Push Notification



SMS-Based OTP



Hardware Tokens



Biometrics



Certificate-Based Authentication



FIDO2 /
Passwordless MFA

MFA in Enterprise Environments

■ Use Cases

- VPN access
- Cloud applications
- Privileged access
- Remote workforce
- Admin accounts

■ Integration Points

- Identity Providers (IdP)
- Active Directory / Azure AD
- SSO platforms
- ZTNA and SASE solutions



That's all Folks!